TASK 1: RESEARCH REPORT – ROBOTICS APPLICATIONS IN HEALTHCARE

TITLE: How surgical robots are changing modern healthcare

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INTRODUCTION:

Robotics has moved from factory floors to hospital operation theatres. A surgical robot is not a robot that operates alone. It is a highly precise machine controlled by a human surgeon through a console. The surgeon sees a magnified 3D view and moves robotic arms that can rotate, bend and cut with more stability than a human hand. This technology started in the late 1980s, but became popular after the da Vinci Surgical System was approved in 2000. Today, surgical robots are used in more than 70 countries for heart, kidney, prostate, gynaecology, and general surgeries. The main goal is not to replace doctors, but to make surgery safer, smaller cuts, less pain, and faster recovery for patients.

How Surgical Robots Work?

A typical system has 3 parts: Surgeon Console: The surgeon sits here, sees a 10x magnified 3D image and controls joysticks. Patient Cart: 3-4 robotic arms with tiny instruments and a camera are placed near the patient. Vision System: A high-definition 3D camera sends live video. When the surgeon moves their hand 1 cm, the robot moves only 1 mm. This is called motion scaling. It also removes hand tremor. The instruments have 7 degrees of freedom, more than a human wrist, so they can reach deep areas through a 1-2 cm cut.

Types of Surgical Robots

Robotic-Assisted Systems: Most common. Example: da Vinci Xi, Hugo. Surgeon fully controls. Supervisory-Controlled Systems: Robot does pre-planned steps, surgeon supervises. Used in hip replacement, example: MAKO. Shared-Control Systems: Surgeon and robot work together. Robot gives force feedback to prevent mistakes.

Applications in Healthcare

Urology: Prostate cancer removal. Robot helps preserve nerves, less blood loss.

Gynaecology: Hysterectomy and fibroid removal with small cuts.

Cardiac Surgery: Heart valve repair without opening full chest.

Orthopaedics: Knee and hip replacement with accurate implant placement. General Surgery: Gallbladder, hernia, colon surgery.

Neurosurgery: Brain tumour biopsy with high accuracy.

Rehabilitation Robots: After surgery, robots like Lokomat help patients walk again.

Advantages:

Smaller incisions (1-2 cm vs 15-20 cm), less pain and scars Less blood loss and infection risk. Shorter hospital stay: 1-2 days vs 5-7 days in open surgery. More precision in complex areas Surgeon can sit comfortably, less fatigue in 5-6 hour surgeries. Better for patients with diabetes, obesity, where healing is slow.

Limitations and Challenges:

Very high cost: One da Vinci system costs Rs 12-15 crore, plus Rs 1-2 lakh maintenance per surgery

Training: Surgeon needs 100+ hours training.

No touch feeling: Surgeon cannot feel tissue, only sees it.

Technical failure risk, though very rare (less than 0.1%).

Not available in small towns and government hospitals in India yet.

Ethical question: Who is responsible if robot makes error?

Robotics in India:

India now has 100+ surgical robot installations, mainly in Apollo, Medanta, AIIMS Delhi, and Manipal hospitals. The Indian government under Make in India is supporting SSI Mantra, an Indian-made surgical robot that costs 1/3rd of da Vinci. This will make robotic surgery affordable for middle-class patients in next 5 years.

Future Scope:

In future, AI will help robots suggest the safest cut, 5G will allow a surgeon in Delhi to operate on a patient in Hyderabad (remote telesurgery), and tiny nanorobots will deliver medicine inside the body without surgery. Surgical robots will not replace surgeons, but surgeons who use robots will replace those who don't.

Conclusion:

Surgical robots have proved that technology and human skill together give the best results. They make surgery safer, faster, and less painful. The challenge is cost and access. As Indian companies make cheaper robots, this technology will reach every district hospital.

References:

Intuitive Surgical - da Vinci Surgical System Official Website

World Health Organization - Medical Devices Report

Indian Journal of Surgery - Robotic Surgery in India, 2023